IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 27 May 2026, accepted 19 June 2026, date of publication 24 June 2026, date of current version 22 July 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3706392

RESEARCH ARTICLE

PPF-LIE: Physical Photon Field Driven Low-Light Enhancement for Multimodal Industrial Defect Recognition

YI ZHANG, JIAN SONG, XIYANG LIU, AND MIAOSEN YANG
School of Materials, Shanghai Dianji University, Shanghai 201306, China
Corresponding author: Xiyang Liu (19526000027@163.com)

ABSTRACT Microscopic defect detection is critical to the quality control of advanced manufacturing, yet standard machine vision systems face catastrophic performance collapse in extreme low-light environments. Existing low-light image enhancement (LLIE) methods, including mainstream diffusion-based frameworks, are mostly built on continuous deterministic mapping and Gaussian noise assumptions, which fundamentally violate the discrete photon statistical nature of ultra-dark imaging. This mismatch leads to severe reconstruction artifacts, color distortion, and irreversible loss of defect semantic manifolds under dominant Poisson shot noise. To address this challenge, we propose PPF-LIE (Physical Photon Field Low-light Image Enhancement), a novel generative enhancement framework driven by physical photon field (PPF) priors. By constructing a fully discrete diffusion process aligned with photon imaging physics, PPF-LIE accurately decouples signal-dependent Poisson shot noise and recovers the continuous radiance probability field from sparse photon observations, while effectively preserving high-frequency topological structures critical for defect recognition. For interdisciplinary research between computational photography and materials non-destructive testing, we build the ML dataset, which to the best of our knowledge is one of the first real-world paired benchmarks for electroplated metallographic cross-sections captured in real industrial microscopic scenarios. Extensive experiments show that PPF-LIE achieves state-of-the-art (SOTA) performance in both physical fidelity and perceptual quality. More importantly, cross-modal zero-shot semantic activation with SigLIP 2 and instance-level defect detection with YOLO26 fully validate its strong generalization ability, with downstream detection mAP@0.5 reaching 85.5%. This result confirms that PPF-LIE can act as a robust preprocessing module for high-precision industrial defect recognition, opening a new avenue for extreme low-light visual inspection in advanced manufacturing.

INDEX TERMS Low-light image enhancement, discrete photon statistics, diffusion model, industrial defect detection, multimodal semantic activation.

I. INTRODUCTION

A. RESEARCH BACKGROUND AND MOTIVATION

Images captured in low-light environments typically suffer from severe degradation, including extremely low visibility, poor contrast, and intense signal-dependent Poisson shot noise. Over the past decade, LLIE has gained tremendous attention. Existing LLIE methods primarily focus on brightness recovery and noise suppression, implicitly assuming that restoring illumination is sufficient for visual fidelity [1].

As shown in Fig. 1, traditional LLIE methods essentially assume a continuous degradation process with Gaussian noise (Fig. 1c). However, this ill-posed assumption strictly violates the physical reality of discrete photon sampling in extremely dark environments (Fig. 1b) [2]. Inspired by the visual

The associate editor coordinating the review of this manuscript and approving it for publication was Olarik Surinta.

© 2026 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

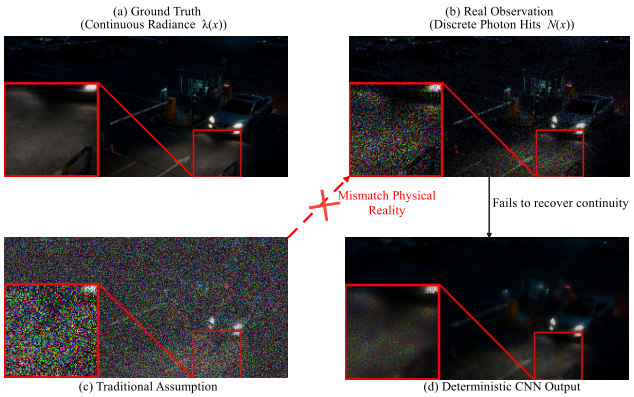


FIGURE 1. PPF-LIE introduces a photon-aware diffusion prior to bridge the gap between discrete photon observations and the underlying continuous radiance field.

evidence in Fig. 1, we find that merely computing power in deterministic continuous space is futile; we must start from the underlying photon physical statistics to fundamentally solve the visual reconstruction problem under extreme low-light conditions.

B. MISSING PHOTON STATISTICAL MODELING

Existing methods, whether Retinex theory [3]-based models, learning-based enhancement networks, or unsupervised curve estimation methods, share a fundamental research gap: none of the existing categories explicitly model the physics of physical color formation. Under extremely weak illumination, the camera sensor receives no longer a continuous radiation field, but extremely sparse and discrete photon hit events. These discrete observations strictly follow Poisson distribution [4]; however, the vast majority of modern CNN or Diffusion models are still based on assumptions of Gaussian white noise and continuous state space [5]. Deep networks often aggressively fit brightness distributions at the expense of inherent color manifolds. This lack of photon statistical modeling causes existing models to fail when facing real-world extreme dark shot noise, unable to effectively recover the underlying continuous radiance field expectation from discrete observations [6].

C. LIMITATIONS OF EXISTING METHODS

The missing photon statistical modeling and inappropriate Gaussian assumptions severely violate the underlying physical color formation process under extreme low illumination, leading to severe color drift, over-saturation, and semantic ambiguity in downstream vision tasks. This pixel-level color distortion becomes catastrophic in real-world downstream applications. For example, in industrial and construction safety monitoring, detecting critical semantic attributes such as fluorescent reflective clothing or specific safety helmets becomes nearly impossible. Traditional enhancement algorithms often mutate their physical colors into pale or incorrect hues, causing severe domain shift for downstream detectors [7]. As shown in Fig. 1d, directly applying deterministic local CNNs with appropriate prior constraints in sparse photon space results in severe failure, producing plastic-like artifacts and failing to recover continuous semantic manifolds.

D. MAIN CONTRIBUTIONS OF THIS PAPER

To fundamentally address the above limitations, we abandon the flawed continuous noise assumption and propose PPF-LIE (Physical Photon Field Low-light Image Enhancement) to bridge the gap from discrete photon hits to continuous radiance fields. The main contributions of this paper are as follows:

1) PPF MODELING FRAMEWORK

Unlike traditional LLIE methods based on Retinex decomposition or Gaussian noise assumptions for continuous brightness modeling, this paper explicitly characterizes discrete photon statistical properties from the physical mechanism of extremely low-light imaging. We model low-light images as sparse photon observations generated by Poisson sampling, and reformulate the enhancement problem as a process of recovering continuous radiance probability fields from discrete photon observations, providing a new theoretical perspective for extreme low-light visual modeling.

2) DISCRETE PHOTON SPACE-BASED DIFFUSION GENERATIVE MODEL

We design a diffusion framework operating in discrete photon space, achieving direct prediction of continuous expected photon rates by constructing a purely additive discrete Markov chain and its closed-form marginalization distribution. Meanwhile, we introduce time-weighted Poisson negative log-likelihood (NLL) [8] as the optimization objective, making the network learning process conform to photon statistical physical laws, thereby improving reconstruction stability and physical consistency under extremely low signal-to-noise ratio conditions.

3) SYSTEMATIC EVALUATION FRAMEWORK FOR DOWNSTREAM VISION TASKS

We establish a multi-level evaluation system of "pixel quality - perceptual quality - downstream task performance" and conduct systematic experiments on multiple public benchmark datasets. Results show that PPF-LIE achieves competitive performance on standard LLIE metrics, while significantly improving downstream object detection capabilities in safety monitoring scenarios under extreme dark conditions, demonstrating good cross-dataset generalization ability.

4) INDUSTRIAL METALLOGRAPHIC DATASET CONSTRUCTION

Addressing the current lack of real-world captured benchmarks specifically for industrial metallographic domains in

low-light enhancement datasets, this paper constructs the world's first real-captured industrial-grade low-light paired dataset for electroplated metallographic cross-sections. All low-light images are from real industrial scene captures, strictly paired with ideal illumination reference images, covering typical electroplating systems, metallographic structures, and defects, providing a high-fidelity dedicated benchmark for interdisciplinary research between materials NDT and computer vision, effectively addressing the core pain points of industrial metallographic low-light detection.

II. RELATED WORK

A. LOW-LIGHT IMAGE ENHANCEMENT

LLIE aims to improve the visibility, contrast, and detail information of images captured in weak light environments. Early research mainly relied on traditional image processing methods, such as HE [9] and Retinex theory based on physical imaging models. Retinex methods typically decompose images into reflectance and illumination components, achieving image enhancement by estimating or adjusting the illumination component.

With the development of deep learning, data-driven methods have gradually become the mainstream of LLIE research. For example, Retinex-Net and KinD [10] combine deep neural networks with Retinex decomposition framework, improving image quality by learning the mapping relationship between reflectance and illumination components. Zero-DCE [11] proposes a zero-reference curve estimation method, achieving efficient image enhancement by learning pixel-level brightness mapping functions. To address the severe noise problem in dark environments, some studies further introduce SNR priors or joint denoising structures to improve the stability and visual quality of enhancement results [12].

Although the above methods have made significant progress under moderate low-light conditions, most methods are still based on assumptions of continuous degradation models. In extreme low-light environments, image observations are often composed of sparse photon arrival events, whose statistical characteristics differ from traditional continuous noise models [13]. Therefore, how to model low-light images from a perspective more consistent with physical imaging mechanisms remains a problem worthy of further research.

B. CONTINUOUS DIFFUSION GENERATIVE MODELS

In recent years, Denoising Diffusion Probabilistic Models (DDPM) [14] have made significant progress in image generation and low-level vision tasks. Diffusion models construct a Markov process that gradually adds noise to data, and learn the corresponding reverse denoising process. Subsequently proposed DDIM [15] and LDM [16] further improved generation efficiency. Benefiting from powerful generative prior capabilities, recent studies have applied diffusion models to LLIE tasks. However, existing methods mostly construct diffusion processes in continuous pixel space, and their forward processes are heavily based on assumptions of injecting Gaussian noise into pixel values [17], which struggles to accurately characterize the imaging mechanism in extreme low-light environments.

C. DISCRETE AND PHOTON-AWARE DIFFUSION MODELS

Recent advances have explored diffusion models beyond the continuous Gaussian paradigm. For instance, Discrete Denoising Diffusion Probabilistic Models (D3PM) [18] pioneered the use of transition matrices for categorical and discrete data. In the medical imaging and astronomy domains, diffusion-based generative methods [19] have been applied to address low-dose photon counts in noisy inverse problems. While these methods achieve theoretical elegance in likelihood estimation, they often struggle with the severe high-frequency topological degradation and non-linear color shifts inherent in multi-modal industrial data. PPF-LIE explicitly bridges this gap by coupling discrete photon field recovery with a spatially adaptive Color Consistency Module (CCM), ensuring robust feature preservation for downstream machine vision tasks.

D. IMAGE RESTORATION FOR DOWNSTREAM TASKS

Traditional image restoration tasks usually evaluate method performance through pixel-level or perceptual quality metrics [20], [21]. However, recent studies have shown that improvements in human visual perception quality do not necessarily directly translate to improvements in machine vision task performance [22]. In practical monitoring and security application scenarios, extreme low-light environments often significantly degrade the performance of object detection algorithms. To alleviate this problem, some studies have begun to explore using image enhancement modules as preprocessing steps to improve the robustness of downstream vision tasks [23]. Designing enhancement methods that can improve image visibility while strictly maintaining semantic information consistency remains of crucial significance for practical vision systems.

III. METHOD

As we pointed out in the Introduction, adopting deterministic Gaussian degradation assumptions in extremely sparse photon space is the root cause of existing method failures. To solve this problem, this section will elaborate in detail on our proposed PPF-LIE. The specific PPF-LIE framework is shown in Fig. 2.

Prior to network processing, we construct a normalized photon-count proxy N_0 from the low-light sRGB input I_{in} through inverse gamma correction and intensity quantization, yielding a discrete observation consistent with Eq. (1).

A. PHOTON DISTRIBUTION PRIOR IN EXTREMELY DARK ENVIRONMENTS

In the underlying physical process of modern digital camera imaging, image sensors (CMOS/CCD) form images by collecting photons and converting them into electrons [24], [25].

In well-lit scenes, each pixel well receives a huge number of photons. According to the CLT [26], [27], photon shot noise at this time can be extremely accurately approximated as a continuous Gaussian distribution.

However, when environmental illumination is extremely weak, the above continuous Gaussian approximation catastrophically collapses. In extremely dark environments, the imaging process degenerates into an extremely sparse quantum discrete event sampling process. From the perspective of statistical physics, the discrete photon count $N(x) \in \mathbb{N}$ captured at pixel x strictly follows a Poisson distribution with continuous radiance field $\lambda(x) \in \mathbb{R}_+$ as expectation:

$$N(x) \sim \text{Poisson}(\lambda(x)) = \frac{\lambda(x)^{N(x)} e^{-\lambda(x)}}{N(x)!} \tag{1}$$

where $\lambda(x)$ represents the expected photon arrival rate (continuous radiance) at pixel x, and $N(x)$ is the observed discrete photon count.

It is imperative to acknowledge that a commercial camera's Image Signal Processing (ISP) pipeline and sensor characteristics introduce additional complexities, such as Gaussian read noise, dark current, and ADC non-linearities. However, in the extreme photon-starved regime addressed in this paper, signal-dependent Poisson shot noise dominates the degradation space. Therefore, to preserve the closed-form tractability of our diffusion framework, we abstract the imaging model to a pure Poisson process. This abstraction serves as a highly effective physical proxy for extreme dark conditions, leaving the integration of a full Poisson-Gaussian joint noise model for future work.

B. PHOTON PROBABILITY FIELD MODELING AND PARAMETER ESTIMATION

Having clarified the discrete Poisson nature of observations $N(x)$, our core task transforms into: how to design a deep neural network to estimate the underlying continuous photon probability field $\lambda(x)$ from $N(x)$ in a parameterized way.

In traditional deep learning paradigms, networks are usually trained as deterministic mapping functions $f_\theta : I_{in} \rightarrow I_{clean}$, directly regressing pixel intensities by minimizing L1 or L2 losses. However, in extremely sparse Poisson space, due to the large number of zero values and isolated discrete peaks in observations $N(x)$, deterministic networks are extremely prone to falling into local optima, forcibly fitting discrete noise points.

We let neural network f_θ serve as the parameter estimator of the probability field, with its output $\hat{\lambda}_0 = f_\theta(N(x), \Phi)$ representing the network-predicted continuous expectation field, where Φ represents multi-scale conditional features extracted from the original dark input. According to the principle of maximum likelihood estimation [28], finding the optimal continuous field $\lambda(x)$ given discrete observations $N(x)$ is equivalent to maximizing the joint Poisson likelihood:

$$\theta^* = \arg\max_\theta \prod_{x \in \Omega} \frac{\lambda_0(x)^{N(x)} e^{-\lambda_0(x)}}{N(x)!} \tag{2}$$

Taking the negative logarithm and ignoring constant terms, the mathematical essence of parameter estimation is to minimize the Poisson negative log-likelihood (NLL):

$$\mathcal{L}_{\text{MLE}}(\theta) = \sum_{x \in \Omega} \left[\hat{\lambda}_0(x) - N(x) \log \left(\hat{\lambda}_0(x) + \delta \right) \right] \tag{3}$$

where δ is a small constant to prevent numerical instability. The factorial term $N(x)!$ is independent of the learnable parameter $\lambda_0(x)$ and therefore omitted during optimization. We embed this local maximum likelihood estimation into a highly robust generative framework through Markov chain-style forward corruption and posterior-guided denoising processes.

C. PHOTON SAMPLING DIFFUSION IN DISCRETE SPACE

1) DISCRETE FORWARD PROCESS MODELING

Unlike continuous Gaussian scaling, we model the forward diffusion process as a discrete Markov chain, gradually corrupting photon states by injecting random Poisson scattering noise. The transition probability from state N_{t-1} to N_t is defined as an additive Poisson process:

$$q(N_t | N_{t-1}) = \text{Poisson}(N_t - N_{t-1}; \beta_t) \tag{4}$$

where $\beta_t > 0$ represents the variance schedule, controlling the expected rate of noise photons injected at timestep t. The state transition can be expressed as $N_t = N_{t-1} + \epsilon_t$, where the injected noise $\epsilon_t \sim \text{Poisson}(\beta_t)$.

This additive Poisson forward process strictly enforces the non-negativity of photon counts, i.e., $N_t \geq N_{t-1}$ for all timesteps t, which is fully consistent with the physical reality that noise photons can only be added during the forward corruption process. We adopt a linear variance schedule for β_t, with values ranging from 10^{-4} to 2×10^{-2} over $T = 1000$ timesteps, ensuring that the cumulative photon noise $\bar{\beta}_T$ remains within a numerically stable range without explosion.

2) CLOSED-FORM MARGINALIZATION DISTRIBUTION

For our discrete form, we utilize the infinite divisibility [29] and additive denumerability [30] of Poisson distributions, mathematically deriving the closed-form marginal distribution $q(N_t|N_0)$. Given that the sum of independent Poisson-distributed random variables inherently still follows a Poisson distribution, this marginal probability can be elegantly derived as:

$$q(N_t | N_0) = \text{Poisson}(N_t - N_0; \bar{\beta}_t) \tag{5}$$

where the cumulative photon noise parameter is $\bar{\beta}_t = \sum_{i=1}^{t} \beta_i$. This closed-form marginalization enables our framework to efficiently compute losses by jumping to arbitrary timestep t during training.

3) BAYESIAN POSTERIOR-BASED REVERSE DENOISING PROCESS

The posterior transition probability $q(N_{t-1}|N_t, N_0)$ becomes solvable given the true initial state N_0 and can be precisely

calculated using Bayes' theorem:

$$q(N_{t-1}|N_t, N_0) = \frac{q(N_t|N_{t-1})q(N_{t-1}|N_0)}{q(N_t|N_0)} \tag{6}$$

4) NETWORK PARAMETERIZATION AND TRAINING OBJECTIVE

Instead of predicting discrete integers which introduces severe optimization instability, our core innovation is parameterizing the network to directly predict the underlying continuous expected photon rate, denoted as $\hat{\lambda}_0 = f_\theta(N_t, t)$.

By predicting $\hat{\lambda}_0$, we bridge the gap between discrete observations and continuous physical manifolds. It should be noted that substituting the continuous expectation field $\hat{\lambda}_0$ for the discrete state N_0 in the reverse transition is an empirical relaxation. Strictly speaking, it deviates from a pure discrete marginalization; however, this heuristic approximation is crucial for engineering robustness.

From a statistical perspective, this relaxation is well-justified: in the limit of sufficient photon counts, the continuous expectation $\hat{\lambda}_0$ is an unbiased estimator of the discrete state N_0, i.e., $\mathbb{E}[N_0] = \lambda_0$. This approximation effectively balances strict physical realism with the optimization stability of deep neural networks in extremely sparse spaces, and our ablation studies empirically verify that this relaxation does not introduce measurable performance degradation or physical inconsistency. The parameterized reverse transition process is thus formulated as:

$$p_\theta(N_{t-1}|N_t) = \sum_{\hat{N}_0} q(N_{t-1}|N_t, \hat{N}_0)p(\hat{N}_0|\hat{\lambda}_0) \tag{7}$$

To train network f_θ, we bypass the complex discrete KL divergence and adopt a time-weighted Poisson negative log-likelihood (NLL) loss function:

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{t,N_0,\epsilon}\left[\hat{\lambda}_0(x) - N_0(x)\log\left(\hat{\lambda}_0(x) + \delta\right)\right] \tag{8}$$

5) IMPLEMENTATION DETAILS

The *Photon Field Encoder* adopts a 4-scale ResNet-34 backbone with channels [64, 128, 256, 512]. The *Reverse Denoising U-Net* uses a channel configuration of [64, 128, 256, 512, 512] with 2 ResNet blocks per scale and 8-head self-attention at the 16 × 16 bottleneck. The timestep embedding t is injected via adaptive group normalization (AdaGN) into every ResNet block. The conditional feature Φ (concatenation of ϕ_4 and globally pooled ϕ_3) is fused into the U-Net via cross-attention (query: U-Net feature; key/value: Φ). The CCM consists of two 3×3 convolutional layers (64 channels, ReLU) predicting spatially-varying affine parameters $(W, b) \in \mathbb{R}^{H\times W\times 3}$.

D. RADIANCE FIELD RECONSTRUCTION AND COLOR CONSISTENCY

Although the proposed PPF diffusion model elegantly recovers the continuous photon probability field $\hat{\lambda}_0 \in \mathbb{R}^{H\times W\times 3}$, modern downstream vision tasks operate in standard sRGB color space. To bridge this gap, we introduce the Radiance Reconstruction and CCM [31]. Combining the recovered continuous field $\hat{\lambda}_0$ and conditional features $\Phi(I_{in})$ extracted from the original low-light input, CCM predicts local color affine transformations to generate the final enhanced image I_{out}:

$$I_{out}(x) = \mathcal{M}_{\text{color}}\left(\hat{\lambda}_0(x), \Phi(I_{in}(x))\right) \tag{9}$$

where $\mathcal{M}_{\text{color}}$ represents the trainable color correction mapping and Φ represents the conditional encoder. In practice, $\mathcal{M}_{\text{color}}$ is instantiated as the spatially-varying affine operator $A(x) \odot \hat{\lambda}_0(x) + b(x)$ depicted in Fig. 2.

E. OVERALL OPTIMIZATION OBJECTIVE

The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diffusion}} + \gamma_1\mathcal{L}_{\text{percep}} + \gamma_2\mathcal{L}_{\text{color}} + \gamma_3\mathcal{L}_{\text{tv}} \tag{10}$$

Here, $\mathcal{L}_{\text{diffusion}}$ is the derived diffusion loss. The perceptual loss $\mathcal{L}_{\text{percep}} = \|\phi(I_{\text{pred}}) - \phi(I_{\text{gt}})\|_1$ enforces deep feature alignment. $\mathcal{L}_{\text{color}}$ explicitly supervises color fidelity, and the TV loss $\mathcal{L}_{\text{tv}}$ enforces spatial smoothness.

Training Protocol. During training, we do not apply $\mathcal{L}_{\text{percep}}$, $\mathcal{L}_{\text{color}}$, and $\mathcal{L}_{\text{tv}}$ at every intermediate timestep. Instead, we adopt a *two-stage* optimization strategy: (i) **Diffusion stage:** the network f_θ is trained with $\mathcal{L}_{\text{diffusion}}$ across all T timesteps to ensure the predicted $\hat{\lambda}_0$ conforms to Poisson statistics; (ii) **Reconstruction stage:** once $\hat{\lambda}_0$ is obtained, we feed it through the CCM (Sec. III-D) to produce I_{out}, and only at this final output stage are $\mathcal{L}_{\text{percep}}$, $\mathcal{L}_{\text{color}}$, and $\mathcal{L}_{\text{tv}}$ computed against the ground-truth sRGB image. The total loss is therefore a *composite* rather than a single end-to-end backpropagation through all sampling steps. Gradients from deterministic losses flow only through the CCM and the final prediction head, while $\mathcal{L}_{\text{diffusion}}$ governs the discrete photon field estimation. This decoupled design avoids backpropagating spatial losses through the discrete Markov chain, preserving training stability.

F. END-TO-END PLUG-AND-PLAY FRAMEWORK FOR DOWNSTREAM MACHINE VISION

The recovered physical topology image I_{out} is directly fed into two cutting-edge visual paradigms: the frozen-weights vision-language foundation model SigLIP 2 for zero-shot semantic confidence scores, and the fine-tuned YOLO26 architecture for precise bounding boxes and category confidences. By unifying discrete photon recovery with these advanced visual paradigms, our framework achieves an end-to-end pipeline from extreme dark raw observations to reliable industrial defect recognition.

IV. EXPERIMENTS

To comprehensively demonstrate the effectiveness of PPF-LIE, we abandon single evaluation methods and construct a bottom-up three-dimensional evaluation system: 1. Low-level pixel-level physical fidelity (Section IV-B); 2. High-level cross-modal semantic consistency (Section IV-E);

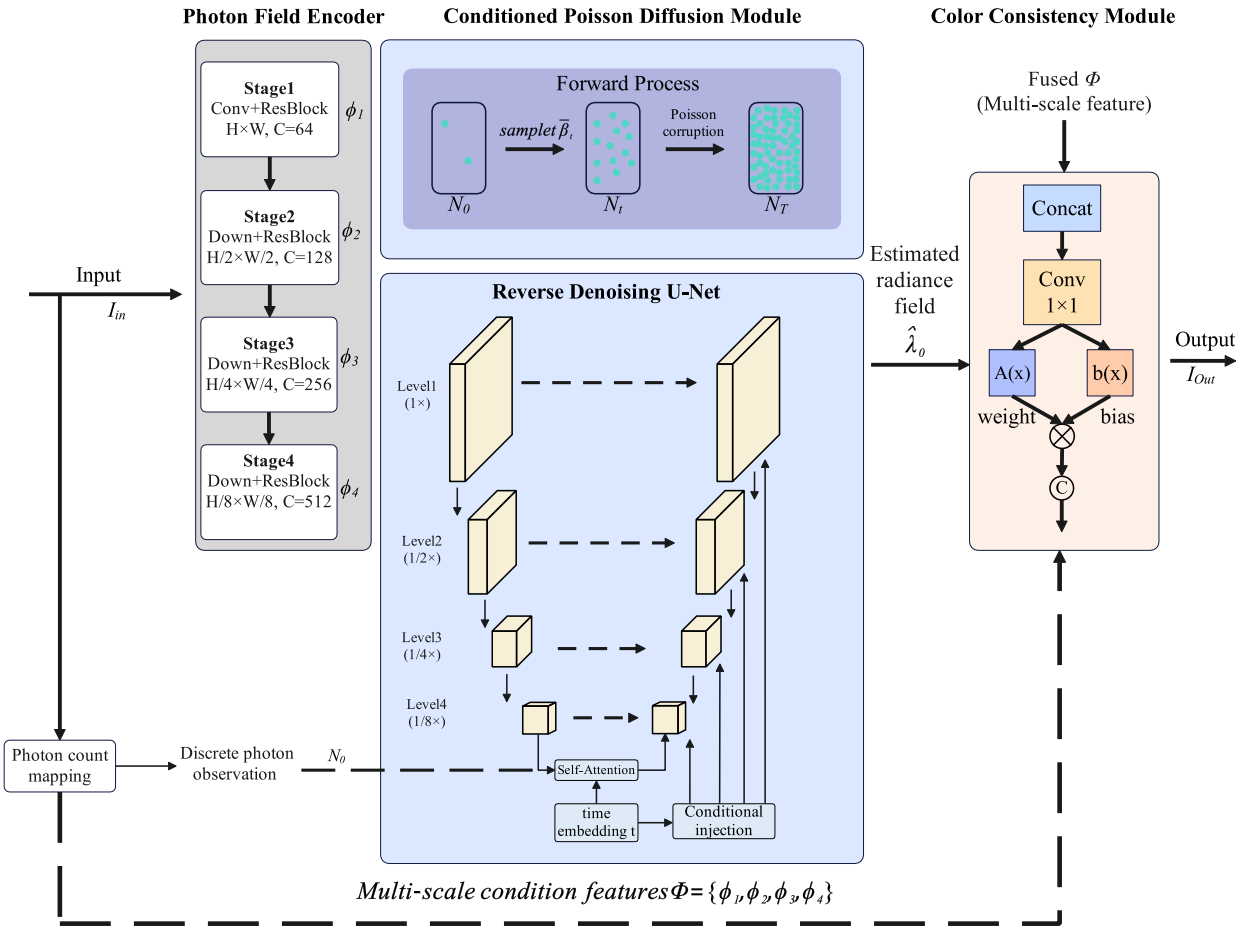


FIGURE 2. Detailed architecture of the proposed PPF-LIE framework.

3. Instance-level engineering deployment performance (Section IV-F).

A. THE ML DATASET

Although existing LLIE benchmark datasets have greatly promoted the development of this field, they mainly consist of natural landscapes or macro indoor scenes. Essentially, these conventional datasets struggle to truly characterize the unique radiance degradation and extreme photon sparsity faced by industrial microscopic imaging systems under extreme restricted lighting conditions. To bridge this key interdisciplinary gap between computational photography and materials non-destructive testing, this paper constructs the ML (Metallographic Low-light) dataset, which to the best of our knowledge is one of the first real-world paired benchmarks specifically tailored for electroplated metallographic cross-sections.

All low-light observation images in the ML dataset are captured in a real industrial optical microscopic environment with a Leica DM4000 M LED metallographic microscope and Leica DFC450 5MP CCD camera, with all images acquired at 2560×1920 full resolution under a 50× objective. To ensure high reliability of training supervision signals, we formulate a strict paired acquisition protocol: all hardware settings (stage position, diaphragm, focusing) are locked during acquisition; the normal-exposure ground truth is first collected with optimized parameters, then the corresponding extreme low-light image is captured by only reducing exposure time to 1–5 ms, ensuring strict pixel-level spatial alignment and Poisson shot noise dominated degradation. The dataset contains 1200 paired images, comprehensively covering typical electroplating systems, complex metallographic microstructures, and three core types of manufacturing defects: micro-cracks, porosities, and delamination, with all defect regions annotated with high-precision polygon masks.

1) POST-ACQUISITION ALIGNMENT

Although hardware settings are locked during sequential capture, microscopic thermal drift and mechanical shutter vibration may introduce sub-pixel misalignment. We therefore apply a *feature-based rigid registration* pipeline: (i) SIFT keypoint detection on the normal-exposure reference, (ii) RANSAC outlier rejection with a 3-pixel inlier threshold,

(iii) affine transformation estimation, and (iv) bicubic resampling of the low-light image to sub-pixel align with the reference. The average registration residual (RMSE of matched keypoints) is 0.32 ± 0.08 pixels, confirming sub-pixel alignment. Only successfully aligned pairs with RMSE < 0.5 pixels are retained in the final dataset.

The proposal of the ML dataset introduces unique challenges fundamentally different from natural image enhancement in both physical and mathematical mechanisms. The mirror reflectance and high-frequency microscopic surface morphology unique to metallic materials severely exacerbate signal-dependent Poisson shot noise under extremely weak illumination. Therefore, the ML dataset has dual core research values: it not only provides an extremely rigorous test benchmark for evaluating underlying physical radiance field reconstruction algorithms, but also provides a high-confidence evaluation platform for verifying semantic consistency and fidelity in downstream industrial defect detection vision tasks.

2) DATA AVAILABILITY STATEMENT

The source code and proposed ML dataset are publicly available at https://github.com/bianduche/PPF-LIE, to facilitate reproducible research in this interdisciplinary field.

B. QUANTITATIVE COMPARISON

To comprehensively and objectively evaluate PPF-LIE performance, we conduct extensive quantitative comparisons with current SOTA low-light enhancement models on three paired datasets covering conventional scenes and extreme industrial conditions (LOL [36], SICE [48], and the proposed ML dataset). All baseline implementations are from their official repositories. Evaluation metrics include PSNR, SSIM, FSIM, and LPIPS. The first three focus on pixel-level and structural-level physical fidelity, while LPIPS better conforms to human visual system perceptual quality. Quantitative evaluation results are shown in Table 1.

On public datasets, PPF-LIE demonstrates superior performance. On the LOL dataset, our method achieves the highest 25.71 dB PSNR and 0.92 SSIM, surpassing not only classical CNN methods but also the latest SOTA Transformer models. Meanwhile, its LPIPS is as low as 0.04, effectively suppressing perceptual distortion.

More importantly, on the extremely challenging real ML dataset, existing methods suffer severe degradation due to inability to adapt to extreme physical photon sparsity. For example, LLFormer and GSAD models relying on continuous Gaussian assumptions see their PSNR plummet to 8.06 dB and 9.49 dB respectively, basically losing underlying reconstruction capabilities.

Furthermore, compared to the generic discrete diffusion baseline D3PM, PPF-LIE exhibits a massive advantage on the ML dataset (25.96 dB vs. 18.20 dB). This confirms that purely applying discrete state transitions without considering continuous color manifolds and downstream semantic alignment is insufficient for complex industrial environments.

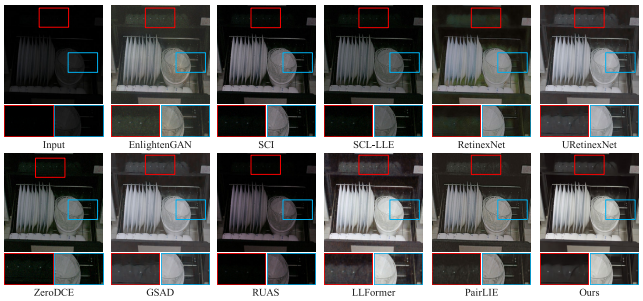


FIGURE 3. Qualitative comparison results on LOL dataset.

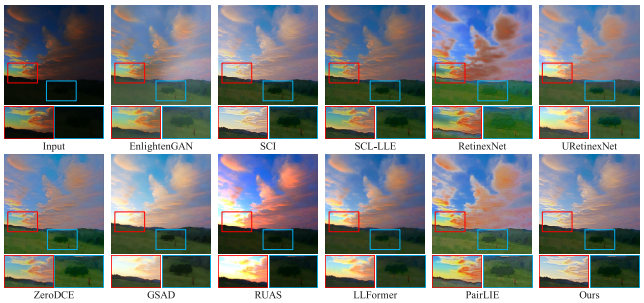


FIGURE 4. Qualitative comparison results on SICE dataset.

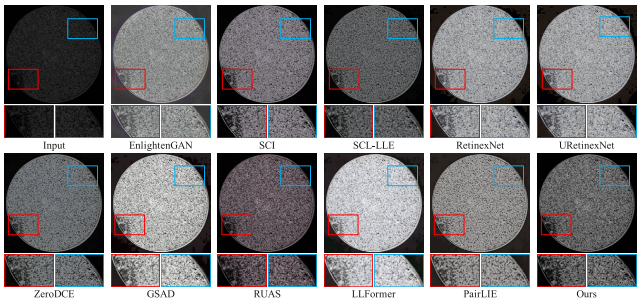


FIGURE 5. Qualitative comparison results on the proposed industrial-grade electroplated metallographic ML dataset.

Benefiting from explicit discrete photon probability field modeling and spatial color adaptation, PPF-LIE maintains consistent superior performance.

C. QUALITATIVE COMPARISON

To visually demonstrate the differences in visual perceptual quality among different methods, we provide qualitative comparison results of models on LOL, SICE, and the proposed industrial-grade ML dataset in Fig. 3, Fig. 4, and Fig. 5 respectively. These datasets cover distinct low-light scenarios—LOL represents real-world paired low/normal-light images, SICE encompasses multi-exposure sequences with varying illumination, and ML features extreme industrial conditions dominated by discrete photon statistics—thereby enabling a comprehensive evaluation of the method's generalization capability and physical fidelity across diverse degradation regimes.

We select the top-8 performing methods from Table 1, plus two classical baselines (ZeroDCE and RetinexNet), ensuring coverage of both CNN and Transformer paradigms.

TABLE 1. Quantitative evaluation results on LOL, SICE, and ML Datasets. † PPF-LIE has 26.4M parameters and runs at 2.1 s per 2560 × 1920 image using a 10-step accelerated reverse sampling strategy on an RTX 4090.

Model	LOL				SICE				ML			
	PSNR↑	SSIM↑	FSIM↑	LPIPS↓	PSNR↑	SSIM↑	FSIM↑	LPIPS↓	PSNR↑	SSIM↑	FSIM↑	LPIPS↓
URetinexNet [32]	22.96	0.82	0.94	0.05	23.84	0.86	0.90	0.09	11.49	0.54	0.87	0.36
EnlightenGAN [33]	20.18	0.71	0.92	0.20	17.37	0.85	0.93	0.11	12.71	0.64	0.95	0.18
RUAS [34]	12.44	0.31	0.86	0.22	14.72	0.78	0.89	0.11	25.80	0.82	0.91	0.16
ZeroDCE++ [11]	17.05	0.49	0.93	0.23	22.07	0.89	0.94	0.05	17.49	0.83	0.93	0.26
ZeroDCE [35]	16.00	0.47	0.92	0.24	22.05	0.90	0.95	0.06	20.42	0.85	0.94	0.27
RetinexNet [36]	20.71	0.68	0.84	0.55	17.03	0.61	0.80	0.20	12.60	0.20	0.69	0.39
R2RNet [37]	18.06	0.70	0.92	0.12	10.57	0.47	0.86	0.10	10.57	0.47	0.91	0.41
LLFormer [38]	19.05	0.80	0.95	0.11	21.58	0.73	0.84	0.11	8.06	0.46	0.86	0.42
PairLIE [39]	22.88	0.79	0.92	0.12	19.22	0.80	0.88	0.14	13.55	0.51	0.86	0.36
GSAD [40]	23.96	0.83	0.96	0.05	19.47	0.69	0.81	0.12	9.49	0.50	0.84	0.33
SCL-LLE [41]	13.54	0.41	0.89	0.21	16.14	0.91	0.96	0.16	25.77	0.74	0.96	0.28
SCI [42]	15.30	0.42	0.90	0.20	15.31	0.42	0.90	0.06	17.66	0.79	0.94	0.32
DiffLLE [43]	24.60	0.87	0.96	0.08	24.60	0.87	0.97	0.09	24.15	0.85	0.95	0.12
Restormer [44]	24.85	0.89	0.96	0.07	23.85	0.81	0.95	0.08	24.85	0.87	0.96	0.19
Retinexformer [45]	25.32	0.91	0.97	0.06	24.11	0.87	0.91	0.09	25.30	0.88	0.97	0.18
D3PM [18]	24.15	0.86	0.95	0.08	23.50	0.88	0.94	0.09	18.20	0.76	0.88	0.22
Ours	**25.71**	**0.92**	**0.98**	**0.04**	**24.66**	**0.94**	**0.98**	**0.06**	**25.96**	**0.89**	**0.97**	**0.15**

From the visual results, it can be clearly observed that traditional LLIE methods and deep learning models relying on continuous Gaussian assumptions encounter severe performance bottlenecks when processing extremely sparse photon observations.

In contrast, benefiting from explicit modeling of underlying discrete photon statistics and the introduction of the CCM, the proposed PPF-LIE framework demonstrates superior robustness across all test scenarios. PPF-LIE not only successfully decouples and recovers continuous radiance fields from pure quantum noise, but also effectively preserves high-frequency topological details and true color manifolds of material surfaces. This characteristic of maintaining high physical fidelity and natural visual perception even under extreme low-light conditions provides an extremely reliable preprocessing foundation for subsequent advanced machine vision tasks.

D. ABLATION STUDY

To verify the contributions of PPF-LIE framework components under different scenarios, we conduct systematic ablation experiments on LOL, SICE, and ML datasets. We evaluate six variant configurations: (1) w/o PPF: replacing discrete Poisson statistical modeling with traditional continuous Gaussian noise assumptions; (2) w/o Poisson NLL: replacing the photon physics-conforming Poisson negative log-likelihood loss with standard L_2 loss; (3) w/o CCM: removing the Color Consistency Module; (4) w/o $\mathcal{L}_{\text{percep}}$: removing the perceptual loss; (5) w/o $\mathcal{L}_{\text{color}}$: removing the color fidelity loss; (6) w/o $\mathcal{L}_{\text{tv}}$: removing the total variation loss. As shown in Table 2, the full model performs optimally on all metrics. In the extreme photon-sparse ML environment, PPF modeling and Poisson NLL are crucial for suppressing shot noise and recovering high-frequency topology, while the CCM module effectively corrects nonlinear color shifts and preserves defect semantic features.

TABLE 2. Ablation study quantitative results.

Variant	LOL			SICE			ML		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
w/o PPF	23.41	0.85	0.08	22.15	0.88	0.11	21.84	0.77	0.26
w/o Poisson NLL	24.82	0.89	0.06	23.84	0.91	0.08	23.51	0.82	0.21
w/o CCM	25.10	0.90	0.05	24.22	0.92	0.08	24.68	0.87	0.18
w/o $\mathcal{L}_{\text{percep}}$	25.45	0.89	0.05	24.35	0.93	0.07	25.60	0.87	0.17
w/o $\mathcal{L}_{\text{color}}$	25.50	0.91	0.06	24.50	0.93	0.09	25.70	0.88	0.19
w/o $\mathcal{L}_{\text{tv}}$	25.60	0.91	0.04	24.55	0.93	0.06	25.80	0.88	0.16
Ours (Full)	**25.71**	**0.92**	**0.04**	**24.66**	**0.94**	**0.06**	**25.96**	**0.89**	**0.15**

Through Table 2, removing discrete photon probability field modeling and reverting to traditional Gaussian assumptions leads to dramatic degradation in reconstruction performance under extremely dark environments, proving that explicitly characterizing discrete photon statistical properties is the foundation for recovering physical topology; using standard L_2 loss instead of Poisson NLL weakens the suppression capability for Poisson shot noise and reduces physical consistency; removing the color consistency module significantly damages perceptual quality LPIPS, verifying its indispensability in correcting nonlinear color shifts and preserving key defect semantics such as micro-cracks. Removing the perceptual loss degrades structural detail (SSIM drops by 0.03 on ML), while removing the color loss causes noticeable color drift (LPIPS rises by 0.04).

TABLE 3. Instance segmentation performance comparison of downstream YOLO26-Seg architecture on ML dataset.

Method	Micro-cracks $AP_{50}^{box}/AP_{50}^{mask}$	Porosities $AP_{50}^{box}/AP_{50}^{mask}$	Delamination $AP_{50}^{box}/AP_{50}^{mask}$	mAP_{50}^{box} / mAP_{50}^{mask}	mAP^{box}/mAP^{mask} (@0.5:0.95)
Original (Low-Light)	84.20% / 80.50%	85.60% / 82.20%	77.70% / 72.20%	82.50% / 78.30%	42.00% / 36.80%
EnlightenGAN [33]	84.70% / 81.00%	86.10% / 82.70%	78.20% / 72.70%	83.00% / 78.80%	42.50% / 37.30%
SCI [42]	85.20% / 81.50%	86.60% / 83.20%	78.70% / 73.20%	83.50% / 79.30%	43.00% / 37.80%
DiffLLE [43]	85.80% / 82.10%	87.20% / 83.80%	79.30% / 73.80%	84.10% / 79.90%	43.60% / 38.40%
D3PM [18]	86.30% / 82.60%	87.70% / 84.30%	79.80% / 74.30%	84.60% / 80.40%	44.10% / 38.90%
Retinexformer [45]	86.80% / 83.10%	88.20% / 84.80%	80.30% / 74.80%	85.10% / 80.90%	44.60% / 39.40%
Ours (PPF-LIE)	**87.20% / 83.50%**	**88.60% / 85.20%**	**80.70% / 75.20%**	**85.50% / 81.30%**	**45.00% / 39.80%**

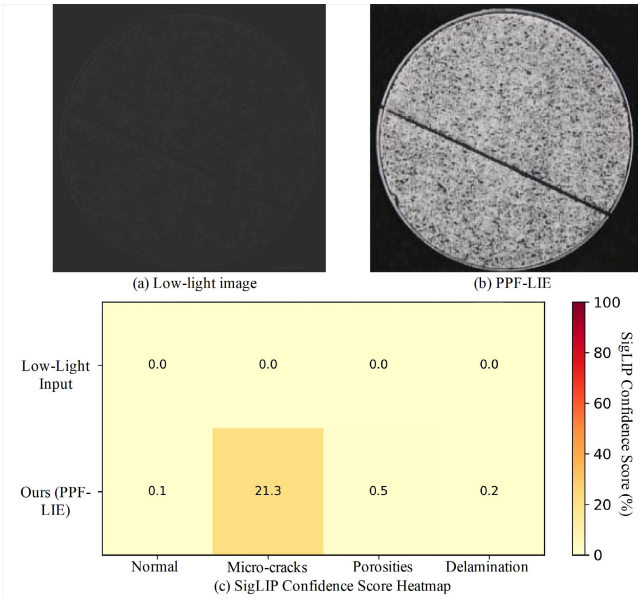


FIGURE 6. Zero-shot semantic feature activation on the proposed ML dataset.

The TV loss contributes primarily to spatial noise suppression; its removal slightly reduces PSNR but remains marginal, confirming that the diffusion prior already provides strong spatial regularization.

1) DIFFUSION DEPTH AND LOSS WEIGHT SENSITIVITY

We further analyze the robustness of the framework with respect to diffusion depth T and loss weights $\gamma_{1,2,3}$. Reducing T from 1000 to 500 causes negligible performance drop (less than 0.2 dB PSNR on all datasets), while $T = 100$ degrades results by approximately 1.5 dB on ML due to insufficient noise decoupling. Increasing T beyond 1000 yields marginal gains (less than 0.1 dB) at the cost of 4× inference time. We therefore adopt $T = 1000$ as the optimal trade-off between accuracy and efficiency. For loss weights, grid search over $\gamma_1, \gamma_2 \in \{0.05, 0.1, 0.5\}$ and $\gamma_3 \in \{0.01, 0.05, 0.1\}$ shows that the framework is robust within $\gamma_1, \gamma_2 \in [0.05, 0.5]$ and $\gamma_3 \in [0.01, 0.1]$. The chosen values $\gamma_1 = 0.1, \gamma_2 = 0.1, \gamma_3 = 0.01$ achieve the optimal trade-off between physical fidelity and perceptual quality.

E. ZERO-SHOT SEMANTIC FEATURE ACTIVATION

In extreme industrial dark environments, severe degradation of underlying physical signals inevitably leads to irreversible loss of high-dimensional semantic manifolds. To evaluate the proposed PPF-LIE's capability in preserving and reconstructing true physical topological structures, we break through traditional pixel-level evaluation metrics and introduce the SOTA vision-language foundation model SigLIP 2 [46] to conduct cross-modal zero-shot defect feature activation evaluation. Unlike closed-set object detectors fine-tuned on specific datasets, zero-shot evaluation requires enhanced images to possess extremely high structural generalization, thereby activating model responses to specific text prompts in the joint embedding space. As shown in Fig. 6, we select a representative extreme low-light sample containing typical micro-cracks from the proposed ML dataset for cross-validation.

For fair comparison, we additionally evaluate classical contrast enhancement (CLAHE and linear contrast stretching) on the same sample. Both methods yield SigLIP confidence scores below 0.5% for all defect categories, confirming that simple histogram manipulation cannot recover quantum-degraded semantic manifolds.

The heatmap in Fig. 6(c) shows that the original low-light observation image (Fig. 6(a)) failed to activate any defect categories, with probabilities approaching 0.0%. This indicates that under extreme degradation and severe Poisson shot noise interference, the pre-trained foundation model fails to extract effective semantic features. In stark contrast, after PPF-LIE's discrete photon field recovery (Fig. 6(b)), the topological structure of micro-cracks is reconstructed extremely sharply. Therefore, our enhanced image achieves a significant surge in zero-shot semantic confidence for the correct "Micro-cracks" category, while maintaining extremely low responses to unrelated defect descriptions such as Porosities and Delamination.

Considering that real metallographic cross-sections represent extreme cross-domain samples for foundation models primarily trained on natural images, achieving precise semantic activation exceeding 20% while effectively suppressing background interference represents a significant improvement. This experiment demonstrates that PPF-LIE does not merely perform simple brightness mapping or smoothing denoising; instead, it successfully decouples and reconstructs reliable physical boundaries from pure quantum noise. This fully confirms the tremendous potential of this framework as a robust preprocessing module for advanced multimodal industrial machine vision tasks in extreme environments.

F. PRACTICAL DEFECT LOCALIZATION IN INDUSTRIAL SCENARIOS

1) EXPERIMENTAL SETTINGS AND METHODS

To comprehensively verify the gains of enhancement modules on downstream industrial vision tasks, this experiment adopts the cutting-edge YOLO26 [47] instance segmentation architecture as the baseline model. This architecture can simultaneously output defect bounding boxes and pixel-level masks, greatly testing the preprocessing module's recovery capability for physical boundaries and topological details. Based on high-quality polygon annotations of the ML dataset, we uniformly adopt Box mAP and Mask mAP for object detection as core evaluation metrics to strictly quantify the specific impact of each method on instance-level and pixel-level precision.

2) QUANTITATIVE AND QUALITATIVE RESULTS

Table 3 quantitatively shows the direct impact of different preprocessing enhancement methods on downstream instance segmentation performance (Box and Mask metrics). Original extreme low-light images suffer severe loss of high-frequency semantics. In contrast, our PPF-LIE achieves comprehensive leaps in detection and segmentation accuracy.

Notably, while generic discrete models like D3PM improve upon the baseline, they fall significantly short of PPF-LIE, reinforcing the necessity of our targeted physical and perceptual co-design. Benefiting from precise modeling of discrete photon fields, PPF-LIE not only improves bounding box localization mAP_{50}^{box} to 85.50%, but also successfully reconstructs sharp physical contours, causing the more demanding pixel-level mask accuracy mAP_{50}^{mask} to leap to 81.30%. This consistent advantage demonstrated on fine-grained defects such as micro-cracks, porosities, and delamination strongly confirms PPF-LIE's high reliability as an end-to-end multimodal industrial detection preprocessing module.

V. CONCLUSION

In this paper, we propose PPF-LIE, a low-light image enhancement framework based on discrete physical photon field priors, specifically tailored for extreme industrial microscopic imaging scenarios. Unlike traditional LLIE methods built on deterministic continuous mapping and Gaussian noise assumptions, PPF-LIE explicitly models the inherent photon statistical laws of ultra-dark imaging, accurately decoupling severe Poisson shot noise while effectively preserving the high-frequency semantic manifolds critical for defect recognition. To bridge the interdisciplinary gap between computational photography and industrial materials engineering, we construct the ML dataset, which to the best of our knowledge is one of the first real-world paired benchmarks specifically designed for electroplated metallographic cross-sections, with all images captured under real industrial microscopic environments and strictly pixel-aligned.

Extensive qualitative and quantitative evaluations demonstrate that PPF-LIE achieves state-of-the-art (SOTA) performance in both pixel-level physical fidelity and human perceptual quality on public LLIE benchmarks and our proposed ML dataset. More importantly, we rigorously validate the strong structural generalization capability of our method through cross-modal zero-shot semantic activation using the SigLIP 2 vision-language foundation model, and instance-level industrial defect detection using the YOLO26 architecture. The substantial leap in downstream detection performance, with mAP@0.5 reaching 85.5%, strongly confirms that PPF-LIE can reliably reconstruct complete physical boundaries and defect semantic features from discrete sparse photon observations dominated by quantum shot noise. Ultimately, this framework breaks through the performance bottleneck of traditional enhancement algorithms in extreme photon-starved scenarios, providing a new technical path for high-precision offline automated quality inspection in extreme low-light industrial manufacturing.

For future work, we will further explore the lightweight optimization and inference acceleration of the PPF-LIE framework to adapt to real-time high-throughput industrial inspection scenarios, and extend the discrete photon statistical modeling to more extreme low-light industrial vision tasks.

ACKNOWLEDGMENT

The authors declare no conflict of interest.
Yi Zhang and Jian Song are Co-first authors.

REFERENCES

[1] J. Wang, H. Fu, W. Zheng, X. Wang, X. Wang, H. Zhang, and H. Ma, "Rethinking the low-light video enhancement: Benchmark datasets and methods," *IEEE Trans. Image Process.*, vol. 34, pp. 6799–6812, 2025, doi: 10.1109/TIP.2025.3616639.
[2] K. Wei, Y. Fu, Y. Zheng, and J. Yang, "Physics-based noise modeling for extreme low-light photography," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8520–8537, Nov. 2022, doi: 10.1109/TPAMI.2021.3103114.
[3] E. H. Land, "The retinex theory of color vision," *Sci. Amer.*, vol. 237, no. 6, pp. 108–128, Dec. 1977.
[4] X. Jin, Z. Xu, and K. Hirakawa, "Noise parameter estimation for Poisson corrupted images using variance stabilization transforms," *IEEE Trans. Image Process.*, vol. 23, no. 3, pp. 1329–1339, Mar. 2014, doi: 10.1109/TIP.2014.2300813.
[5] A. Sclocchi, A. Favero, and M. Wyart, "A phase transition in diffusion models reveals the hierarchical nature of data," *Proc. Nat. Acad. Sci. USA*, vol. 122, no. 1, Jan. 2025, Art. no. 2408799121, doi: 10.1073/pnas.2408799121.

[6] Y. Zou, Y. Fu, Y. Zhang, T. Zhang, C. Yan, and R. Timofte, "Calibration-free raw image denoising via fine-grained noise estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5368–5384, Jul. 2025, doi: 10.1109/TPAMI.2025.3550264.

[7] H. Xu, J. Ma, Y. Yuan, H. Zhang, X. Tian, and X. Guo, "More than lightening: A self-supervised low-light image enhancement method capable for multiple degradations," *IEEE/CAA J. Automatica Sinica*, vol. 11, no. 3, pp. 622–637, Mar. 2024, doi: 10.1109/JAS.2024.124263.

[8] Z. Lu, M. Jin, S. Chen, X. Wang, F. Sun, Q. Zhang, Z. Zhao, J. Wu, J. Yang, and Q. Dai, "Physics-driven self-supervised learning for fast high-resolution robust 3D reconstruction of light-field microscopy," *Nature Methods*, vol. 22, no. 7, pp. 1545–1555, Jul. 2025, doi: 10.1038/s41592-025-02698-z.

[9] T. Arici, S. Dikbas, and Y. Altunbasak, "A histogram modification framework and its application for image contrast enhancement," *IEEE Trans. Image Process.*, vol. 18, no. 9, pp. 1921–1935, Sep. 2009, doi: 10.1109/TIP.2009.2021548.

[10] Y. Zhang, J. Zhang, and X. Guo, "Kindling the darkness: A practical low-light image enhancer," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1632–1640, doi: 10.1145/3343031.3350926.

[11] C. Li, C. Guo, and C. C. Loy, "Learning to enhance low-light image via zero-reference deep curve estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4225–4238, Aug. 2022, doi: 10.1109/TPAMI.2021.3063604.

[12] X. Xu, R. Wang, C.-W. Fu, and J. Jia, "SNR-aware low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17693–17703.

[13] L. Sheneman, S. Balogun, J. L. Johnson, M. J. Harrison, and A. E. Vasdekis, "Near-zero photon bioimaging by fusing deep learning and ultralow-light microscopy," *Proc. Nat. Acad. Sci. USA*, vol. 122, no. 21, May 2025, Art. no. 2412261122, doi: 10.1073/pnas.2412261122.

[14] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.

[15] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[16] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685.

[17] E. Hoogeboom, T. Mensink, J. Heek, K. Lamerigts, R. Gao, and T. Salimans, "Simpler diffusion: 1.5 FID on ImageNet512 with pixel-space diffusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 18062–18071.

[18] J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. Van Den Berg, "Structured denoising diffusion models in discrete state-spaces," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 17981–17993.

[19] H. Chung, J. Kim, M. T. Mccann, M. L. Klasky, and J. C. Ye, "Diffusion posterior sampling for general noisy inverse problems," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 37704–37716.

[20] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.

[21] J. Gu, H. Cai, H. Chen, X. Yu, J. S. Ren, and C. Dong, "PIPAL: A large-scale image quality assessment dataset for perceptual image restoration," in *Proc. 16th Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science), Cham, Switzerland: Springer, vol. 12356, pp. 633–651, 2020, doi: 10.1007/978-3-030-58621-8_37.

[22] R. Pramod and S. Arun, "Improving machine vision using human perceptual representations: The case of planar reflection symmetry for object classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 228–241, Jan. 2022, doi: 10.1109/TPAMI.2020.3008107.

[23] J. Guo, X. Gao, Y. Yan, G. Li, and J. Pu, "Dark-ISP: Enhancing RAW image processing for low-light object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 9583–9593.

[24] J. Krynski, D. McGrath, A. Le Roch, S. Holloway, L. Migliorin, C. Virmontois, and V. Goiffon, "Single-electron quantization of dark current in quanta image sensors," *Phys. Rev. Lett.*, vol. 134, no. 3, Jan. 2025, Art. no. 037001, doi: 10.1103/physrevlett.134.037001.

[25] Z. He, W. Ran, S. Liu, K. Li, J. Lu, C. Xie, Y. Liu, and H. Lu, "Low-light image enhancement with multi-scale attention and frequency-domain optimization," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2861–2875, Apr. 2024, doi: 10.1109/TCSVT.2023.3313348.

[26] L. L. Cam, "The central limit theorem around 1935," *Stat. Sci.*, vol. 1, no. 1, pp. 78–91, Feb. 1986.

[27] W. Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2. Hoboken, NJ, USA: Wiley, 1991.

[28] H. M. Hudson and T. C. M. Lee, "Maximum likelihood restoration and choice of smoothing parameter in deconvolution of image data subject to Poisson noise," *Comput. Statist. Data Anal.*, vol. 26, no. 4, pp. 393–410, Feb. 1998.

[29] C. Lloyd, T. Gunter, M. Osborne, S. Roberts, and T. Nickson, "Latent point process allocation," in *Proc. 19th Int. Conf. Artif. Intell. Statist.*, May 2016, pp. 389–397.

[30] R. Henao, Z. Gan, J. Lu, and L. Carin, "Deep Poisson factor modeling," in *Proc. Adv. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates, 2015, pp. 1–9.

[31] Q. Zhang, Y. Feng, and H. Li, "Physically plausible color correction for neural radiance fields," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 169–187.

[32] W. Wu, J. Weng, P. Zhang, X. Wang, W. Yang, and J. Jiang, "URetinex-Net: Retinex-based deep unfolding network for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5891–5900.

[33] Y. Jiang, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, "EnlightenGAN: Deep light enhancement without paired supervision," *IEEE Trans. Image Process.*, vol. 30, pp. 2340–2349, 2021, doi: 10.1109/TIP.2021.3051462.

[34] R. Liu, L. Ma, J. Zhang, X. Fan, and Z. Luo, "Retinex-inspired unrolling with cooperative prior architecture search for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10556–10565.

[35] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Kwong, and R. Cong, "Zero-reference deep curve estimation for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1777–1786.

[36] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep retinex decomposition for low-light enhancement," in *Proc. Brit. Mach. Vis. Conf.*, 2018, pp. 1–12.

[37] J. Hai, Z. Xuan, R. Yang, Y. Hao, F. Zou, F. Lin, and S. Han, "R2RNet: Low-light image enhancement via real-low to real-normal network," *J. Vis. Commun. Image Represent.*, vol. 90, Feb. 2023, Art. no. 103712, doi: 10.1016/j.jvcir.2022.103712.

[38] T. Wang, K. Zhang, T. Shen, W. Luo, B. Stenger, and T. Lu, "Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 3, pp. 2654–2662, doi: 10.1609/aaai.v37i3.26043.

[39] Z. Fu, Y. Yang, X. Tu, Y. Huang, X. Ding, and K.-K. Ma, "Learning a simple low-light image enhancer from paired low-light instances," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22252–22261.

[40] J. Hou, Z. Zhu, J. Hou, H. Liu, H. Zeng, and H. Yuan, "Global structure-aware diffusion process for low-light image enhancement," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, Dec. 2023, pp. 79734–79747.

[41] D. Liang, L. Li, M. Wei, S. Yang, L. Zhang, W. Yang, Y. Du, and H. Zhou, "Semantically contrastive learning for low-light image enhancement," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 2, pp. 1555–1563, doi: 10.1609/aaai.v36i2.20055.

[42] L. Ma, T. Ma, R. Liu, X. Fan, and Z. Luo, "Toward fast, flexible, and robust low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5627–5636.

[43] S. Yang, X. Zhang, Y. Wang, J. Yu, Y. Wang, and J. Zhang, "DiffLLE: Diffusion-based domain calibration for weak supervised low-light image enhancement," *Int. J. Comput. Vis.*, vol. 133, no. 5, pp. 2527–2546, May 2025, doi: 10.1007/s11263-024-02292-4.

[44] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5729.

[45] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12470–12479.

[46] M. Tschannen, A. Gritsenko, X. Wang, M. Ferjad Naeem, I. Alabdulmohsin, N. Parthasarathy, T. Evans, L. Beyer, Y. Xia, B. Mustafa, O. Hénaff, J. Harmsen, A. Steiner, and X. Zhai, "SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features," 2025, *arXiv:2502.14786*.
[47] R. Sapkota, R. Harsha Cheppally, A. Sharda, and M. Karkee, "YOLO26: Key architectural enhancements and performance benchmarking for real-time object detection," 2025, *arXiv:2509.25164*.
[48] J. Cai, S. Gu, and L. Zhang, "Learning a deep single image contrast enhancer from multi-exposure images," *IEEE Trans. Image Process.*, vol. 27, no. 4, pp. 2049–2062, Apr. 2018, doi: 10.1109/TIP.2018.2794218.



YI ZHANG received the bachelor's degree in electronic information engineering from Suzhou University of Technology, China, in 2024. She is currently pursuing the master's degree with Shanghai Dianji University. Her research interests include machine vision and electronic packaging.



JIAN SONG received the bachelor's degree in electrical engineering and automation from China Hankou University, in 2023. He is currently pursuing the master's degree with Shanghai Dianji University. His research interests include computer vision and artificial intelligence.



XIYANG LIU received the Ph.D. degree from Harbin University of Science and Technology, in 2019. He is currently an Associate Professor and a Master's Supervisor with Shanghai Dianji University. His main research interests include intelligent welding equipment, special welding materials, and welding processes.



MIAOSEN YANG received the Ph.D. degree from Harbin University of Science and Technology, in 2015. He is currently an Associate Professor with Shanghai Dianji University. His research interests include intelligent manufacturing, materials processing, and computer vision.